

Business Scenario	Recommended Approach	Why?
Customer churn prediction for monthly planning	Batch	Historical data is aggregated over time; real-time insights aren't necessary
Real-time fraud detection on transactions	Stream	Requires instant analysis to identify suspicious activity as it happens
Generating daily sales summary reports	Batch	Data is processed at regular intervals; latency isn't critical
Live monitoring of industrial IoT sensors	Batch	Immediate response is needed to detect anomalies or system failures in real time.
Training ML models using months of data	Batch	Involves large historical datasets processed periodically to improve model accuracy
Delivering personalized	Stream	Needs immediate user behavior tracking to deliver timely and relevant suggestions

Figure 3: When to use what?

Table 2: Open source tools for batch and stream processing

Tool	Type	Description	Best use cases	Special notes
 APACHE HADOOP	Batch	Distributed file-based batch processing using MapReduce	Large-scale ETL, log processing, historical analysis	Foundation of Big Data era; used less today
 APACHE SPARK	Batch/Unified	Fast, in-memory processing engine with rich APIs	Batch analytics, ML pipelines, graph processing	Supports micro-batching as well
 APACHE AIRFLOW	Batch	Workflow scheduler for authoring and monitoring complex pipelines	Scheduling ETL jobs, managing DAGs, dependency handling	Widely used in orchestration layer
 Wigzi	Batch	Lightweight Python-based workflow orchestrator from Spotify	Smaller ETL jobs, batch control flows	Simpler alternative to Airflow
 aws Glue	Batch	Serverless ETL platform running on Spark	Scalable jobs within AWS ecosystem	Managed service, pay-per-use model
 kafka	Stream	Distributed pub/sub platform for real-time event streaming	Event ingestion, decoupling services, log processing	Backbone of many real-time data platforms
 Kafka Streams	Stream	Lightweight Java library for processing Kafka streams	Real-time transformations, windowed aggregations	Easy for JVM-based apps
 Apache Flink	Stream/Unified	Stream-first engine with event-time semantics and stateful computation	Fraud detection, complex event processing, alerts	Very low latency; supports batch mode too
 APACHE STORM	Stream	Early low-latency distributed stream processor	Simple stream jobs, legacy systems	Replaced by Flink in most new use cases
 Apache Beam	Unified	Unified model to build batch/stream pipelines; runs on Flink, Spark, Dataflow	Cross-platform pipeline portability	One codebase for both batch and stream
 debezium	Stream	CDC tool built on Kafka Connect to capture database changes	Real-time replication, data sync between systems	Often paired with Kafka for DB streaming
 DELTA LAKE	Unified	Storage layer enabling ACID transactions on top of data lakes (Spark-based)	Combining streaming and batch data in data lakes	Popular in Medallion architecture